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TIPPECC Summer School
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Introduction to Basics of Linux

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Shell

- **What is a shell?**

  command-line interface

  Examples:

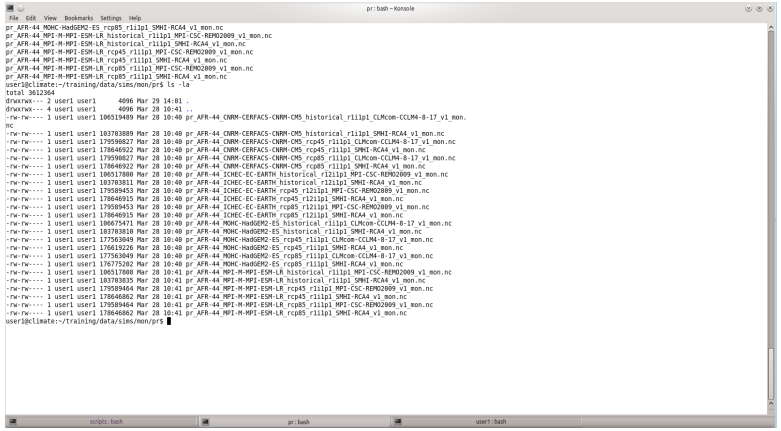
  Bash (Bourne Again Shell): Default on most Linux systems

  Zsh (Z Shell): Default on modern macOS

  PowerShell: Modern Windows shell (older: Command Prompt (cmd.exe))

  ….

- Bash tutorial: http://www.hypexr.org/bash_tutorial.php, www.linuxconfig.org/bash-scripting-tutorial



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Use of shell

Important commands (case-sensitive!)

Each command has a build-in help-function: *command* --help

- Start a program:
  `<program>`, e.g. `firefox`
- Show the content of a directory:
  `ls` or use `ls -l (without hidden files) or ll (with hidden files)` for more details
- Show the current path of the directory:
  `pwd`
- Change a directory:
  `cd <directory>`
- Leave current folder and go back to the parent directory:
  `cd ..`
- Create a new directory:
  `mkdir <directory>`

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Use of shell

Important commands (case-sensitive!)

- Copy one or more files:
  `cp <file> <target_path>`, example: `cp temp.nc /data/precipitation`
- Move a file or a directory:
  `mv <file1> <file2>`
- Delete a file:
  `rm <file>`
- Delete a directory:
  `rm -r <directory>` (recursive: all files in the directory will be removed)
- Start a text editor and open a script file:
  `kate script.sh (vi, nano,.. if you have no ui)`
- Start a script in a directory:
  `./script.sh`
- Make a script executable:
  `chmod +x script.sh` (type `ll` to see execution rights)

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Use of shell

Useful programmes

- Show the content of a text file:
  `more file.txt` (scroll with space key)

- Display one figure:
  `display figure.png`

- Open a figure explorer to have a look at more than one figures:
  `geeqie`

- Open a PDF document:
  `okular file.pdf`

- To use the terminal window after executing a command, put a & at the end of command line before executing the command:
  Example: `okular file.pdf&`

If you don't want to type long filenames manually, you can use the copy- and paste-function. Double-click with the left mouse-button at the filename and press the right mouse-button to get the copy- and paste-menu.

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Use of shell

Practice

1. What is your actual path?
   Hint: Use the command *pwd*!

2. Change in the *~/training* directory and write “This is a climate course!” in a text file named as *test.txt*.
   Hint: Use the command cd and the editor *kate*!

3. Check the content of the text file using the command *more*!

4. Delete the file *test.txt* using the command *rm*!

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Thank you for your attention!

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